Lab Experiment - 6

Aim

To implement a basic N-gram language model using a Bigram Model in Python to generate text based on the given training text.

Algorithm

1. Start the program.
2. Define a sample training text.
3. Split the text into words.
4. Create bigrams by pairing consecutive words.
5. Store the next word for each current word.
6. Select a starting word and generate text using the bigram model.
7. Display the generated text.
8. Stop the program.

Python Program

```
import random

text = "i love python i love programming python is easy"

words = text.split()
bigrams = {}

for i in range(len(words) - 1):
    current = words[i]
    next_word = words[i + 1]

    if current not in bigrams:
        bigrams[current] = []

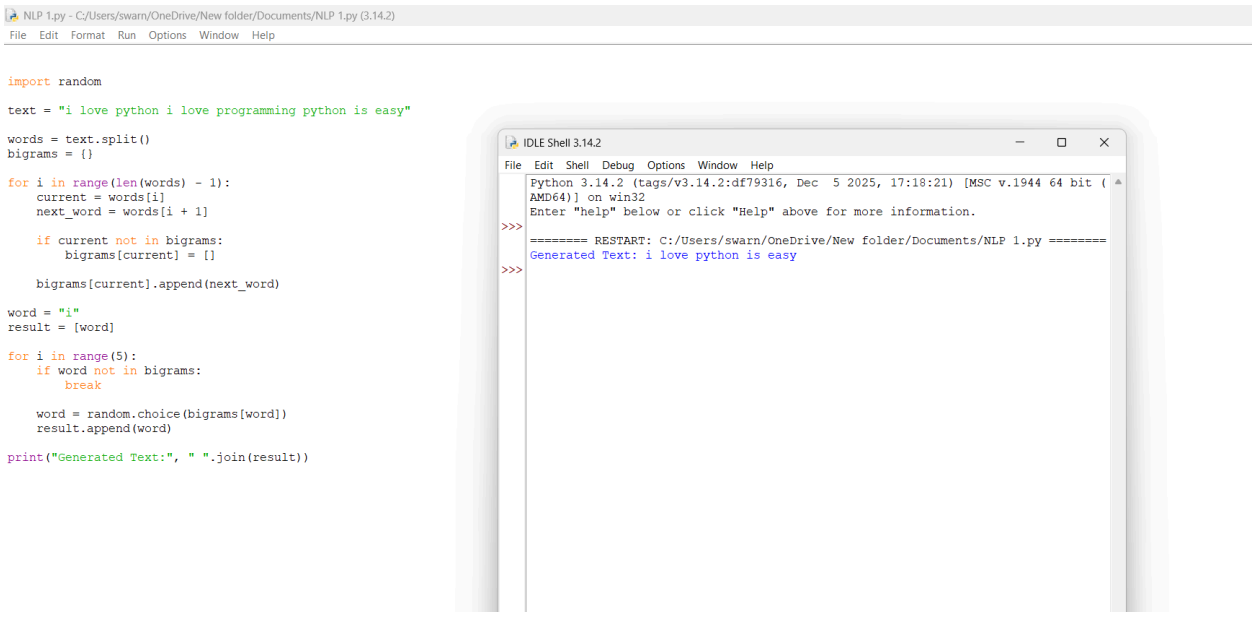
    bigrams[current].append(next_word)

word = "i"
result = [word]
```

```
for i in range(5):
    if word not in bigrams:
        break

    word = random.choice(bigrams[word])
    result.append(word)

print("Generated Text:", " ".join(result))
```



Result

Thus, a basic Bigram N-gram language model was implemented successfully in Python to generate text using consecutive word pairs.